Detailed Design Document: DrowsyGuard DMS

Real-Time AI Driver Drowsiness, Fatigue & Microsleep Detection System | MASTER_SDLC_AGENTS

1. System Architectural Overview

DrowsyGuard DMS is an enterprise-grade automotive safety telemetry engine that operates on the Detective vs. Painter architectural pattern. The perception engine (src/detector.py) isolates facial landmark geometry and temporal transitions, while the presentation tier (src/app.py) composites cockpit HUD graphics, Base64 emergency sirens, and RFC 4180 CSV telemetry logs.

2. The 4-Step Driver Drowsiness Detection Pipeline

- **Step 1: Find Driver's Head:** BlazeFace Deep Learning Locator performs rapid bounding-box extraction (<2ms) across cabin illumination variations.
- **Step 2: Place 3D Digital Mask:** MediaPipe 3D FaceMesh regresses 468 precise facial landmark coordinates tracking eye contours and lip margins.
- **Step 3: Measure Eyes and Mouth:** Computer Vision algorithms compute Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and Perspective-n-Point 3D Head Pose.
- **Step 4: The Smart Stopwatch:** Temporal sliding-window counter distinguishes natural human blinks (2-14 frames) from true microsleeps (>=45 frames).

3. Deep Learning vs. Computer Vision Algorithms Matrix

Algorithm	Domain Type	Input Data	Output Telemetry	Computational Complexity
BlazeFace	Deep Learning (CNN)	Raw RGB Frame (H, W, 3)	Face Bounding Box	O(H x W)
MediaPipe FaceMesh	Deep Learning (CNN)	Cropped Face Image	468 (X, Y, Z) Landmarks	O(N) CNN Inference
Eye Aspect Ratio (EAR)	Computer Vision (Geometry)	6 Eye Landmarks	Scalar Ratio (0.0 to 0.60)	O(1) (6 Points)
Mouth Aspect Ratio (MAR)	Computer Vision (Geometry)	8 Mouth Landmarks	Scalar Ratio (0.0 to 1.50)	O(1) (8 Points)
SolvePnP (Levenberg-Marquardt)	Computer Vision (Matrix Opt)	6 2D Points + 3D Model	Pitch, Yaw, Roll (Degrees)	O(1) Iterative SVD
The Smart Stopwatch	Temporal Logic	Aspect Ratios + Time	ATTENTIVE/DROWSY/CRITICAL	O(1) Frame Accumulator

4. Comparative Backbones: MediaPipe vs. Dlib vs. YOLO-Pose vs. ResNet

Backbone Architecture	Model Family	Landmarks	CPU Latency	Edge Feasibility
MediaPipe FaceMesh (Selected)	Depthwise 3D CNN	468 points	8 - 12 ms	Exceptional (Optimal for Edge)
Dlib Shape Predictor	Gradient Boosted Trees	68 points	25 - 45 ms	Moderate (Fails on Head Slump)
YOLOv8-Pose / YOLO11	CSPDarknet Head	5 - 17 keypoints	30 - 65 ms	Moderate (Coarse Eye Margins)
End-to-End ResNet-50	Heavy CNN Classifier	None (Black Box)	80 - 150 ms	Poor (High Latency, Inscrutable)

5. Real-World Automotive Edge-Case Resolutions

- **Eyeglasses & Lens Reflections:** EAR tracks eyelid margin skin contours rather than pupil center; lens glare does not distort eyelid distance.
- **Driver Talking vs. Yawning:** Speech creates rapid bursts (<0.2s); yawning requires sustained vertical opening (MAR >= 0.65 for >= 12 frames).
- **Night Driving & Low Illumination:** BlazeFace dynamic range normalization combined with near-infrared (NIR) cabin cameras maintains tracking.
- **Looking Down at Navigation / Mirrors:** Head pitch < -25 deg suppresses siren unless eyes are also shut (EAR < 0.22), eliminating false alarms.